# CP255 - W4_S2 Pandas and Dictionaries Answer Key

## Pandas Notebook:

Your Turn: Series Practice

1. **Create a Pandas Series** with the following data:
    - Values: ['apple', 'banana', 'cherry', 'date']
    - Index: ['a', 'b', 'c', 'd']
2. **Display the Series.**
3. **Retrieve a value using .loc[]:**
    - Use .loc to get the value at index "c".
4. **Retrieve a value using .iloc[]:**
    - Use .iloc to get the **third** item in the Series.
5. **Change the index labels** from ['a', 'b', 'c', 'd'] to ['A1', 'B1', 'C1', 'D1'].

```python
fruit_series = pd.Series(
  ['apple', 'banana', 'cherry', 'date'],
  index=['a', 'b', 'c', 'd']
)

fruit_series

print(fruit_series.loc['c'])
print(fruit_series.iloc[2])

fruit_series.index = ['A1', 'B1', 'C1', 'D1']

fruit_series
```

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## Dictionaries Notebook:

Your Turn: Create a dictionary called zoning_codes that maps zoning abbreviations to their descriptions:

"R1" → "Single Family Residential"

"R2" → "Two Family Residential"
"C1" → "Neighborhood Commercial"
"M1" → "Light Industrial"

```python
zoning_codes = {
  "R1": "Single Family Residential",
  "R2": "Two Family Residential",
  "C1": "Neighborhood Commercial",
  "M1": "Light Industrial"
}
```

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Your Turn: Given this list of parcels, create a dictionary that counts how many parcels are in each zoning category:

```python
parcels = [
  {"id": "P001", "zoning": "R1", "acres": 0.5},
  {"id": "P002", "zoning": "R1", "acres": 0.75},
  {"id": "P003", "zoning": "C1", "acres": 1.2},
  {"id": "P004", "zoning": "R2", "acres": 0.6},
  {"id": "P005", "zoning": "R1", "acres": 0.8},
  {"id": "P006", "zoning": "C1", "acres": 2.0},
]

# Create zoning_counts dictionary
zoning_counts = {}

for parcel in parcels:
  zone = parcel['zoning']
  # Your code here - hint: use .get() with a default of 0
  zoning_counts[zone] = zoning_counts.get(zone, 0) + 1

print(zoning_counts)
# Expected: {'R1': 3, 'C1': 2, 'R2': 1}
```